

Notes: Brynjolfsson, Li, and Raymond (QJE, 2025)

Generative AI at Work

Contents

1 Big picture

2 Data and measurement (key objects)

- 2.1 Institutional setting and the AI tool
- 2.2 Rollout and treatment variation
- 2.3 Sample coverage
- 2.4 Main outcome variables
- 2.5 Summary statistics
- 2.6 Skill, tenure, and adherence
- 2.7 Learning, topic, and language measures
- 2.8 Experience-of-work outcomes

3 Models and empirical specifications (all equations + interpretation)

- 3.1 Baseline difference-in-differences specification
- 3.2 Event-study specification
- 3.3 Heterogeneity specifications
- 3.4 Chat-level specifications for outages and topics
- 3.5 How the same regression logic is reused
- 3.6 How to interpret the magnitudes

4 Identification and instruments (what makes the estimates credible)

- 4.1 Why the design is plausible
- 4.2 Core identifying assumption
- 4.3 Why agent fixed effects are essential
- 4.4 Why tenure controls are essential

4.5	Evidence that supports the design	
4.6	Instrumental-variables robustness	
4.7	What remains imperfect	
4.8	Important limitations	
5	Tables and figures	
5.1	Table I and Figure I: the sample and the raw distributions	
5.2	Table II and Figure II: the main productivity result	
5.3	Table III: speed, throughput, and quality	
5.4	Figures III and IV: heterogeneity by skill and tenure	
5.5	Figure V: moving down the experience curve faster	
5.6	Figure VI and Figure VII: adherence and worker learning	
5.7	Figure VIII and Figure IX: topics, fluency, and communication	
5.8	Table IV and Figure X: the experience of work	
6	Short summary	
7	Conclusion	
7.1	Core conclusion (what the paper establishes)	
7.2	Economic intuition	
7.3	Takeaway	

1 Big picture

- **Question.** What happens when a generative-AI assistant is deployed in a real workplace? The paper studies how AI affects worker productivity, who benefits, whether workers learn from the tool, and whether AI changes the day-to-day experience of work.
- **Setting.** The authors study 5,172 chat-based customer-support agents at a Fortune 500 software company. Agents help U.S. small-business customers solve technical problems. The AI system is a GPT-3-based conversational assistant that monitors the chat and suggests replies and relevant documentation.
- **Why this setting is powerful.** Customer support is a task where productivity is measurable, conversations are fully digitized, workers vary widely in skill and experience, and tacit know-how matters a lot. This makes it a good setting to study whether AI transfers knowledge or simply automates routine work.
- **Main empirical design.** The paper exploits staggered individual-level adoption of the AI assistant. Its main design is a difference-in-differences framework with agent fixed effects, year-month fixed effects, and tenure controls, supported by Sun-and-Abraham event studies and several robustness checks.
- **Main average result.** Access to AI raises productivity by about **0.30 resolutions per hour**, which is roughly a **15%** increase relative to the pretreatment mean.
- **Where the gain comes from.** AI mainly makes workers faster and better able to multitask. Average handle time falls by about **3.7 minutes** and chats per hour rise by about **0.365**. Resolution rates rise only slightly, and surveyed customer satisfaction is basically unchanged on average.
- **Strong heterogeneity.** The gains are much larger for lower-skill and less-experienced workers. The bottom skill quintile improves by about **36%**, while the highest-skill workers see almost no productivity gain. Workers with less than one month of tenure gain the most; workers with more than one year of tenure gain little.
- **Interpretation.** The core interpretation is that AI helps diffuse the tacit practices of top performers. Lower-skill workers benefit because the model surfaces techniques and phrasing they were not using before, while top workers have less to learn from the system.
- **Learning rather than pure dependence.** Using data on AI-system outages, the authors show that previously treated workers still handle chats faster even when the AI temporarily stops providing suggestions. That pattern is strongest for workers who adhered more to AI recommendations, which supports a learning channel.
- **Experience-curve effect.** AI compresses ramp-up time. Workers with AI from the start reach the productivity level of much more experienced untreated workers after only a short period on the job.
- **Topic heterogeneity.** The largest gains are not on the easiest or the rarest topics. Instead, AI is most valuable for *moderately rare* problems, where the model still has enough training data but workers have relatively little personal experience.

- **Communication and work experience.** AI improves English fluency, especially for workers in the Philippines, improves customer sentiment, and reduces customer requests to speak to a manager.
- **Broad contribution.** The paper argues that generative AI in this setting behaves more like a *knowledge-transfer and augmentation technology* than a simple replacement technology.
- **Main caveat.** The evidence comes from one firm, one AI tool, and one occupation in the short to medium run. The paper is very informative about local treatment effects in this setting, but not about all occupations or long-run equilibrium effects.
- **Bottom line.** In this workplace, generative AI substantially raises average productivity, narrows within-occupation skill gaps, accelerates learning, and improves some dimensions of the worker experience.

2 Data and measurement (key objects)

2.1 Institutional setting and the AI tool

The data come from a Fortune 500 company that sells business-process software to small and medium-sized businesses in the United States. Customer-support agents answer chat-based technical questions from U.S. customers. These conversations are often complex and last about **40 minutes** on average, with much of the time spent diagnosing the underlying issue.

The AI system is built on **GPT-3** and fine-tuned on historical customer-agent chats labeled with outcomes such as call resolution and handling time. The vendor also up-weights training examples from top performers, so the system is intentionally designed to reflect especially effective worker behavior.

The tool produces two main outputs:

- **real-time response suggestions**, and
- **links to internal technical documentation** relevant to the problem at hand.

The AI is designed to **augment** the worker, not replace them. Agents can ignore, revise, or accept the suggestions.

2.2 Rollout and treatment variation

The broad deployment happened after a small randomized pilot. At scale, rollout was staggered because of:

- limited training capacity,
- a contractual cap on the number of AI-enabled seats,
- time-zone constraints for a global workforce,
- and managers' attempts to avoid service disruptions by spreading workers across sessions.

Agents received AI access after a **three-hour online training session**. These sessions were run by only two AI-firm employees, so adoption timing depended heavily on trainer capacity and scheduling. Crucially, rollout occurred at the **individual level**. Workers in the same team and location often got access at different times. The paper notes that in October 2020 the average team had only about **5%** of active workers using AI, while by January 2021 that figure had risen to about **70%**.

2.3 Sample coverage

The core sample contains:

- **3,006,395 chats**,
- **5,172 agents**,
- **133 teams**,
- and **25 locations**.

Most agents are outside the United States, mainly in the **Philippines**. The paper reports that only about **11%** of agents are U.S.-based.

The post-treatment sample includes about **1.18 million chats** by **1,636 agents**.

2.4 Main outcome variables

The paper studies several standard contact-center outcomes, mostly aggregated to the **agent-month** level.

Average handle time (AHT). Minutes per chat. Lower is better.

Chats per hour (CPH). Number of chats handled per hour, capturing multitasking.

Resolution rate (RR). Share of chats resolved successfully.

Resolutions per hour (RPH). The main productivity outcome: successful issues resolved per hour. Conceptually,

$$\text{RPH}_{it} \approx \text{CPH}_{it} \times \text{RR}_{it}. \quad (\text{A})$$

This is the omnibus productivity measure because it rewards both speed and success.

Net promoter score (NPS). A post-chat survey measure of customer satisfaction.

A practical limitation is that the paper observes the full call-quality data only for a subset of workers, because some subcontractors do not consistently record all quality metrics.

2.5 Summary statistics

For the full sample, the main descriptive moments are approximately:

- average chats per month: **128**,
- average handle time: **41 minutes**,
- standard deviation of handle time: **23 minutes**,

- resolution rate: **0.82**,
- resolutions per hour: **2.1**,
- and customer satisfaction (NPS): about **79**.

The treated group is somewhat stronger than the never-treated group even before rollout, which is why the paper relies heavily on within-worker designs rather than raw cross-sectional comparisons.

2.6 Skill, tenure, and adherence

Worker skill index. The authors build a pre-AI skill index using three components: handling speed, resolution performance, and customer satisfaction. For each company-month they rank workers on these components, average the ranks, smooth using the previous three months, and then place workers into quintiles.

Worker tenure. Experience is measured as months employed when AI is introduced.

AI adherence. The vendor tracks whether the worker follows a suggestion by copying the text or entering a highly similar version. Adherence is the share of generated suggestions that are followed. Average adherence is about **38%**, with an interquartile range of roughly **23%–50%**.

2.7 Learning, topic, and language measures

Outages. To study learning, the paper identifies periods in which the AI system fails to provide suggestions because of technical outages. During major outages, the share of post-treatment chats with no AI suggestions jumps close to 100%.

Topics. The paper uses **Gemini** to classify conversations into topic categories. The topic distribution is highly skewed: payroll/tax issues and account-access/management issues alone make up about **50%** of chats.

Language proficiency. Gemini is also used to score each chat on:

- **comprehensibility** (1 to 5), and
- **native fluency** in American English (1 to 5).

Textual similarity. To study communication change more broadly, the paper computes embeddings of conversation text and tracks cosine similarity over time.

2.8 Experience-of-work outcomes

Customer and agent sentiment. Sentiment scores are built using **SiEBERT** on a scale from -1 to 1 .

Manager escalation requests. The authors measure the share of chats in which the customer asks to speak to a manager or supervisor.

Attrition. In the appendix, the paper also studies turnover responses to AI adoption.

3 Models and empirical specifications (all equations + interpretation)

3.1 Baseline difference-in-differences specification

The main regression is:

$$y_{it} = \delta_t + \alpha_i + \beta AI_{it} + \gamma X_{it} + \varepsilon_{it}. \quad (1)$$

Here:

- y_{it} is an outcome for agent i in month t ,
- AI_{it} equals 1 after the worker receives access,
- δ_t are year-month fixed effects,
- α_i are agent fixed effects,
- and X_{it} mainly includes fixed effects for months of tenure.

Interpretation of β . This is the within-agent change in the outcome after AI access, net of common time shocks and the worker’s normal experience profile. In the preferred RPH regression, β is about **0.301**, which the paper reads as a **15.2%** productivity increase.

3.2 Event-study specification

Because treatment timing is staggered, the paper studies dynamics using the interaction-weighted estimator of Sun and Abraham (2021). A stylized version is:

$$y_{it} = \alpha_i + \delta_t + \sum_{k \neq -1} \beta_k \mathbf{1}\{\text{event time} = k\}_{it} + \gamma X_{it} + \varepsilon_{it}. \quad (2)$$

The omitted period is the month just before adoption. The coefficients β_k trace out pre-trends and post-treatment dynamics.

Interpretation. Flat pre-treatment coefficients support parallel trends. An immediate and persistent post-treatment jump suggests that AI has a fast and durable effect. The event study for RPH in the paper shows exactly that pattern.

3.3 Heterogeneity specifications

To study who benefits, the paper interacts AI access with skill groups, tenure groups, adherence groups, and topic groups. A generic version is:

$$y_{it} = \alpha_i + \delta_t + \sum_g \beta_g (AI_{it} \times G_{ig}) + \gamma X_{it} + \varepsilon_{it}. \quad (3)$$

Here G_{ig} indicates a group such as a skill quintile or a tenure bin.

Interpretation. Each β_g is the treatment effect for group g . These regressions are the heart of

the paper’s distributional message because the average effect hides huge gains for novices and weak performers and almost no gain for top workers.

3.4 Chat-level specifications for outages and topics

Some outcomes are observed at the **chat level**, especially in the learning and topic analyses. A generic version is:

$$y_{ict} = \alpha_i + \delta_t + \beta AI_{it} + \sum_h \theta_h (AI_{it} \times Topic_{ich}) + \gamma X_{ict} + u_{ict}. \quad (4)$$

Here c indexes chats and $Topic_{ich}$ captures either overall topic rarity or the worker’s own familiarity with the topic.

Interpretation. These interactions test whether AI matters most where the model has the most training data, where the human worker has the least experience, or where both conditions are balanced. The paper finds the strongest gains in the middle: moderately rare topics and topics that are relatively unfamiliar to the specific worker.

3.5 How the same regression logic is reused

The same identification framework is then reused for:

- AHT, CPH, RR, and NPS,
- customer and agent sentiment,
- manager-escalation requests,
- language scores,
- and some attrition analyses.

The reading rule is always the same: compare the same worker before and after access, net out common time shocks, and control for tenure.

3.6 How to interpret the magnitudes

The paper reports coefficients in **levels** and then scales them by pretreatment means.

- 0.301 extra resolutions per hour is about **15.2%**.
- -3.746 minutes on AHT is about **8.5%** faster handling.
- 0.365 extra chats per hour is about a **15%** increase in throughput.
- 0.177 on customer sentiment is large because that is about **half a standard deviation** in the paper’s sentiment metric.

4 Identification and instruments (what makes the estimates credible)

4.1 Why the design is plausible

The paper’s identification comes from rollout timing created by operational constraints rather than by the econometrician. The most persuasive elements are:

- limited trainer capacity,
- a hard cap on AI-enabled seats,
- time-zone constraints,
- and worker-level rollout within the same office and team.

This is not perfect randomization, but it creates useful variation in treatment timing that is plausibly unrelated to one worker’s month-to-month productivity shocks.

4.2 Core identifying assumption

The main assumption is a standard **parallel-trends** assumption: absent AI access, treated observations would have evolved like untreated observations once worker fixed effects, time fixed effects, and tenure controls are included.

For the event studies, the paper additionally needs **no anticipation**: workers should not change behavior before access because they know treatment is coming.

4.3 Why agent fixed effects are essential

Eventually treated workers are stronger than never-treated workers even before treatment. That means simple cross-sectional comparisons would overstate the effect of AI. Agent fixed effects absorb permanent worker differences such as baseline talent, language ability, and long-run workplace assignment.

4.4 Why tenure controls are essential

Customer-support workers naturally improve with experience. If AI is introduced later in a worker’s career, raw gains could partly reflect ordinary learning-by-doing. Fixed effects for months of tenure are therefore critical for separating AI from the normal experience curve.

4.5 Evidence that supports the design

Event studies. The main event study shows little evidence of strong pre-trends and a sharp, persistent post-treatment jump.

Alternative staggered-adoption estimators. The paper shows similar results using de Chaisemartin and D’Haultfoeuille, Sun and Abraham, Callaway and Sant’Anna, Borusyak-Jaravel-Spiess,

and traditional two-way fixed effects.

Alternative clustering and weighting. Results are robust to clustering at the worker, team, or location level and to weighting by number of chats.

Pilot RCT. The small randomized pilot before the full rollout produces similar productivity effects.

4.6 Instrumental-variables robustness

The paper’s main design is *not* an IV design. The core strategy is fixed-effects difference in differences.

Still, the appendix reports IV robustness checks that instrument individual adoption timing using broader organizational rollout timing, such as first adoption within the worker’s company, office, or team. Those IV results are similar on many outcomes and somewhat larger for resolutions per hour, suggesting the preferred fixed-effects estimate may be conservative.

4.7 What remains imperfect

The biggest remaining concern is that managers had some discretion over who got trained first. If they systematically prioritized workers with unusually good short-run prospects, the estimates could still be biased upward.

The paper addresses this concern in several ways, but it cannot eliminate it completely.

4.8 Important limitations

- **External validity.** One firm, one product environment, one occupation, one AI tool.
- **Partial equilibrium only.** The paper does not identify long-run effects on wages, staffing, or labor demand.
- **Attrition results are weaker.** Attrition happens once per worker, so those regressions cannot include worker fixed effects.
- **Potential ratchet effects.** Productivity gains may not map one-for-one into permanent compensation gains if firms later raise performance targets.

Overall, the design is very credible for short-run within-firm treatment effects even though it is not a fully randomized rollout at scale.

5 Tables and figures

5.1 Table I and Figure I: the sample and the raw distributions

Read:

- Table I shows a very large operational data set: just over 3 million chats and 5,172 agents.
- Post-treatment workers average about **2.5 resolutions per hour**, compared with about **2.0** for

TABLE I
SUMMARY STATISTICS FOR THE SAMPLE OF CUSTOMER-SERVICE AGENTS

Variable	All	Never treated	Treated, pre	Treated, post
Chats	3,006,395	944,848	881,101	1,180,446
Agents	5,172	3,517	1,340	1,636
Number of teams	133	111	80	81
Share U.S. agents	0.11	0.15	0.081	0.072
Distinct locations	25	25	18	17
Average chats per month	128	83	147	188
Average handle time (min.)	41	43	43	35
St. average handle time (min.)	23	24	24	22
Resolution rate	0.82	0.78	0.82	0.84
Resolutions per hour	2.1	1.7	2	2.5
Customer satisfaction (NPS)	79	78	80	80

Notes. This table shows summary statistics of conversations, agent characteristics, and issue resolution rates, customer satisfaction and average call duration. The first column consists of all agents in our sample, the second column includes control agents who never receive AI access. The third column presents statistics for treated agents before they receive AI access, and the fourth column includes treated agents after AI model deployment.

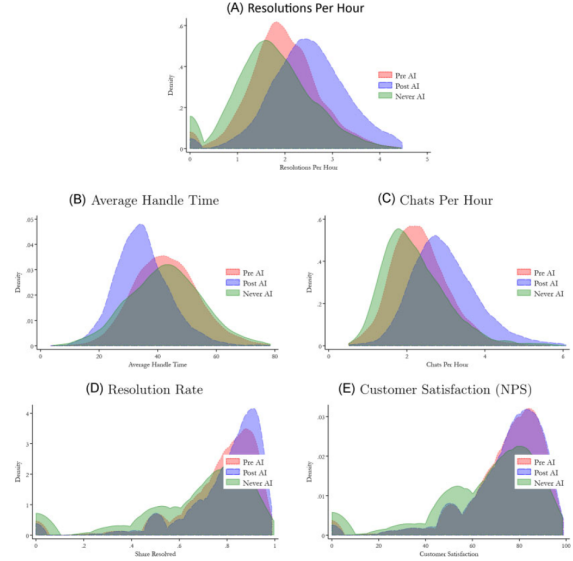


FIGURE I
Raw Productivity Distributions, by AI Treatment

the same workers before treatment and about **1.7** for never-treated workers.

- Average handle time falls from roughly **40 minutes** for untreated or pretreatment observations to about **35 minutes** post treatment.
- Figure I shows that post-treatment workers look better on most raw performance measures.

Economic message. The raw patterns already point toward large gains from AI, but they also show selection. Eventually treated workers were somewhat stronger even before rollout, which is why the fixed-effects design matters so much.

5.2 Table II and Figure II: the main productivity result

TABLE II
MAIN EFFECTS: PRODUCTIVITY (RESOLUTIONS PER HOUR)

Variables	Resolutions/hour (1)	Resolutions/hour (2)	Resolutions/hour (3)
Post AI × Ever treated	0.469*** (0.0325)	0.371*** (0.0318)	0.301*** (0.0329)
Ever treated	0.110** (0.0440)		
Observations	13,192	12,295	12,295
R-squared	0.249	0.562	0.575
Year month FE	Yes	Yes	Yes
Location FE	Yes	Yes	Yes
Agent FE	—	Yes	Yes
Agent tenure FE	—	—	Yes
DV mean	2.123	2.176	2.176

Notes. This table presents the results of difference-in-difference regressions estimating the effect of AI model deployment on our main measure of productivity, resolutions per hour, the number of technical support problems resolved by an agent per hour (resolutions/hour). Post AI × Ever treated captures the impact of AI model deployment on resolutions per hour. Column (1) includes agent geographic location and year-by-month fixed effects. Columns (2) and (3) include agent-level fixed effects, and column (3), our preferred specification described by equation (1), also includes fixed effects that control for months of agent tenure. Observations for this regression are at the agent-month level and all standard errors are clustered at the agent level. Section IV.A describes the AI rollout procedure. Robust standard errors are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

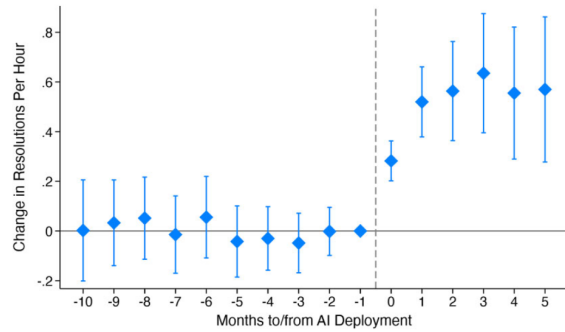


FIGURE II
Event Studies, Productivity

Read:

- Table II, column (1) gives a large raw difference-in-differences estimate of about **0.469** extra

TABLE III
MAIN EFFECTS: ADDITIONAL OUTCOMES

Variables	AHT (1)	Chats/hour (2)	Res. rate (3)	NPS (4)
Post AI $\times$ Ever treated	-3.746*** (0.369)	0.365*** (0.0345)	0.0132 (0.00882)	-0.119 (0.524)
Observations	21,839	21,839	12,295	12,541
R-squared	0.591	0.563	0.371	0.526
Year month FE	Yes	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes	Yes
Agent tenure FE	Yes	Yes	Yes	Yes
DV mean	40.64	2.559	0.822	79.59

Notes. This table presents the results of difference-in-difference regressions estimating the effect of AI model deployment on additional measures of productivity and agent performance. Post AI $\times$ Ever treated measures the impact of AI model deployment after deployment on treated agents for average handle time (AHT) in column (1); chats per hour, the number of chats an agent handles per hour in column (2); resolution rate, the share of technical support problems they can resolve in column (3); and net promoter score (NPS), an estimate of customer satisfaction in column (4). Our regression specification, [equation \(1\)](#), includes fixed effects for each agent, chat year-month, and agent months of tenure. Observations for this regression are at the agent-month level and all standard errors are clustered at the agent level. [Section IV.A](#) describes the AI rollout procedure. Robust standard errors are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

resolutions per hour.

- After adding worker fixed effects and tenure controls, the preferred estimate becomes about **0.301** extra resolutions per hour.
- Relative to the mean of 2.176, this is about a **15.2%** increase in productivity.
- Figure II shows that the increase happens immediately after rollout and remains persistent.

Economic message. The central result is not just that treated workers are better than untreated workers. The same worker becomes materially more productive once AI access begins.

5.3 Table III: speed, throughput, and quality

Read:

- AI lowers average handle time by about **3.746 minutes**.
- AI raises chats per hour by about **0.365**.
- AI raises resolution rates by only about **1.3 percentage points**, and that estimate is imprecise in the main specification.
- AI has essentially **no average effect** on NPS.

Economic message. The productivity gain mainly comes from speed and multitasking rather than from a dramatic increase in average call quality. The average worker does the job faster without obvious deterioration in customer-survey outcomes.

5.4 Figures III and IV: heterogeneity by skill and tenure

Read:

- Figure III shows that the bottom skill quintile gains about **0.5 RPH**, or around **36%**, while the highest-skill quintile gains very little.
- Lower-skill workers improve on AHT, chats per hour, and some quality margins.

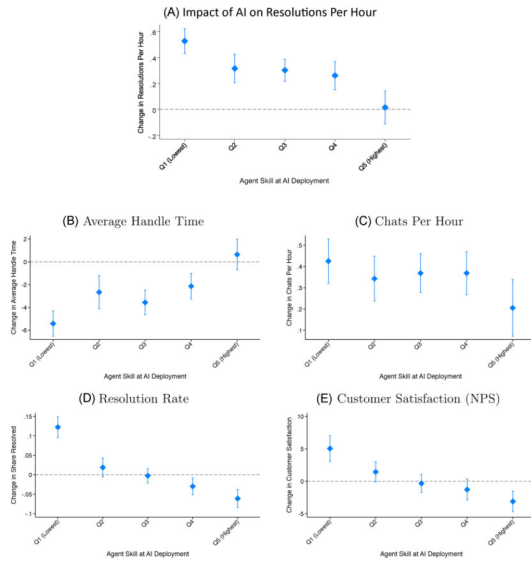


FIGURE III
Heterogeneity of AI Impact, by Skill at Deployment

These figures plot the effects of AI model deployment on five measures of productivity and performance, by pre-deployment worker skill controlling for agent tenure. Agent skill is calculated as the agent's trailing three-month average of performance on average handle time, call resolution, and customer satisfaction, the three metrics our firm uses for agent performance. In each month and company, agents are grouped into quintiles, with the most productive agents in each firm in quintile 5 and the least productive in quintile 1. Panel A plots the effects on resolutions per hour. Panel B plots the average handle time or the average duration of each technical support chat. Panel C graphs chats per hour, or the number of chats an agent can handle per hour. Panel D plots the resolution rate, and Panel E plots net promoter score, an average of surveyed customer satisfaction. All specifications include fixed effects for the agent, chat year-month, and months of tenure. Robust standard errors are clustered at the agent level. The regression specifications are available in [Online Appendix J.C](#) and results in [Online Appendix Table A.VI](#).

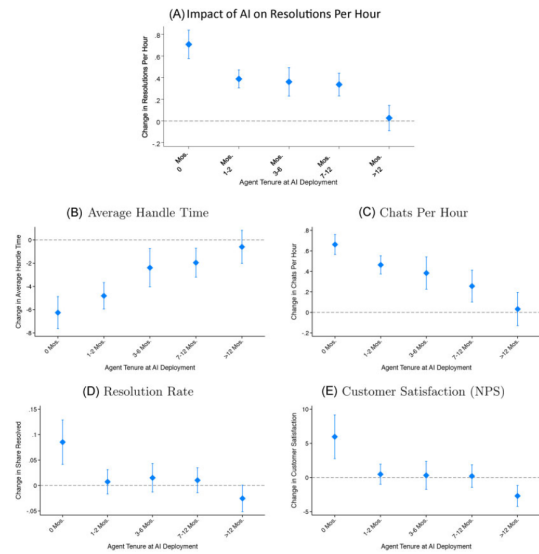


FIGURE IV
Heterogeneity of AI Impact, by Tenure at Deployment

These figures plot the effects of AI model deployment on five measures of productivity and performance by pre-AI worker tenure, defined as the number of months an agent has been employed when they receive access to the AI model. Panel A plots the effects on resolutions per hour. Panel B plots the average handle time or the average duration of each technical support chat. Panel C graphs chats per hour, or the number of chats an agent can handle per hour. Panel D plots the resolution rate, and Panel E plots net promoter score, an average of surveyed customer satisfaction. All specifications include fixed effects for the agent, chat year-month, and months of tenure. Robust standard errors are clustered at the agent level. The regression specifications are available in [Online Appendix J.C](#) and results in [Online Appendix Table A.VI](#).

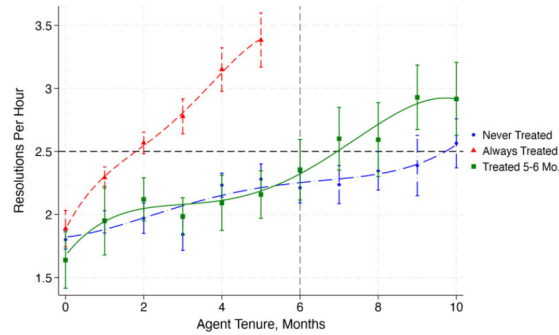


FIGURE V
Experience Curves by Deployment Cohort

This figure plots the relationship between productivity and job tenure. The short-dashed line plots the performance of always-treated agents, those who have access to AI assistance from their first month on the job. The long-dashed line plots agents who are never treated. The solid line plots agents who spend their first four months of work without the AI assistance, and gain access to the AI model during their fifth month on the job. Ninety-five percent confidence intervals are shown. Observations are at the agent-month level.

- Figure IV shows a very similar pattern by experience: the least experienced workers gain the most, while workers with more than a year of tenure gain little.
- Some quality measures for the strongest workers even move slightly downward.

Economic message. Generative AI in this setting is not skill-biased in the classic direction. It narrows within-job performance gaps by helping weaker and newer workers far more than top workers.

5.5 Figure V: moving down the experience curve faster

Read:

- Never-treated workers start near **1.8 RPH** and only reach about **2.5 RPH** after roughly **8–10 months**.
- Workers with AI from the beginning reach about **2.5 RPH** after **only two months** and exceed **3 chats per hour** after around five months.
- Workers treated in month 5 initially follow the untreated path, then bend upward after access and start tracking the always-treated path.

Economic message. AI reduces ramp-up time and steepens the learning curve. It changes not just the level of productivity, but also how quickly workers accumulate effective experience.

5.6 Figure VI and Figure VII: adherence and worker learning

Read:

- Figure VI shows that average adherence is only about **38%**; workers reject many AI suggestions.
- Even the lowest-adherence group gains around **10%** in productivity, but the highest-adherence group gains close to **25%**.

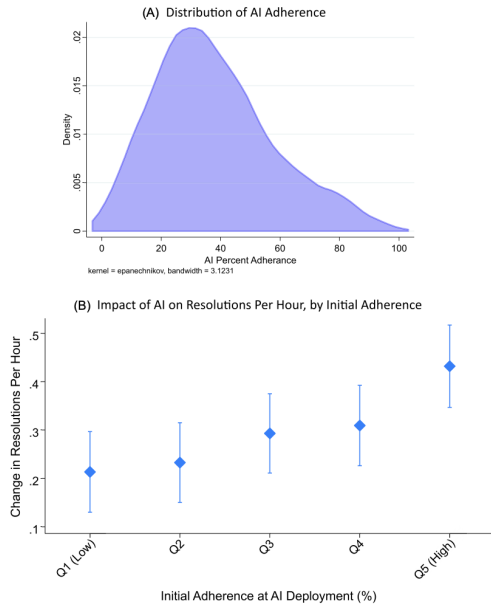


FIGURE VI
Heterogeneity of AI Impact, by AI Adherence

Panel A plots the distribution of AI adherence, averaged at the agent-month level, weighted by the log of the number of AI recommendations for that agent-month. Panel B shows the effect of AI assistance on resolutions per hour, by agents grouped by their initial adherence, defined as the share of AI recommendations they followed in the first month of treatment. The regression, outlined in [Online Appendix J.C](#), is run at the agent-month level and includes fixed effects for agent, chat year-month, and agent tenure in months. Standard errors are clustered at the agent level. Results are in [Online Appendix Table A.VII](#).

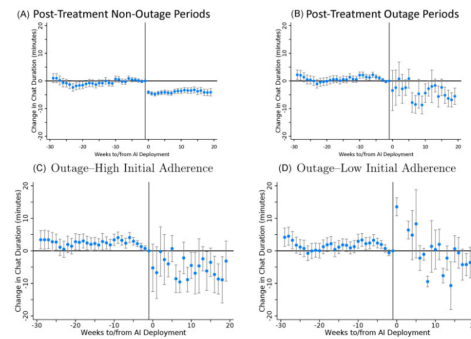


FIGURE VII
Chat Duration during AI System Outages

These figures plot event studies for the effect of AI system rollout of chat duration at the individual-chat level. Panel A restricts to posttreatment chats that do not occur during any period where there is a AI system outage. Panel B restricts to posttreatment chats that only occur during a large system outage. Panels C and D focus on outage only post-periods. Panel C restricts to only chats generated by ever-treated agents who with high initial AI adherence (top tercile), while Panel D restricts to agents with low initial adherence (bottom tercile). Agents who are never treated are excluded from this analysis. The regressions are run at the chat level with agent, year-month, and tenure fixed effects with standard errors clustered at the agent level. Regression results are reported in [Online Appendix Table A.VIII](#).

- Figure VII shows that during major outages, previously treated workers still handle chats about **15%–25%** faster than their own pre-AI baseline if they have had enough exposure.
- These outage-period gains are strongest for workers who initially adhered more to the AI system.

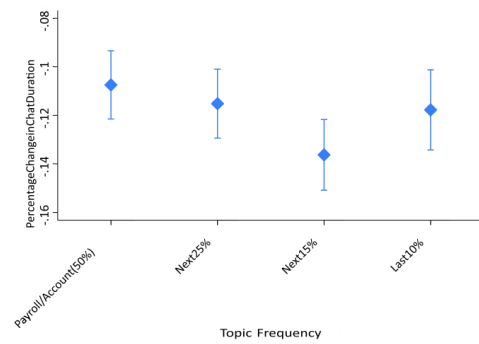
Economic message. This is some of the most important evidence in the paper. AI is not merely a live crutch. Workers seem to internalize useful practices and perform better even when the tool is temporarily unavailable.

5.7 Figure VIII and Figure IX: topics, fluency, and communication

Read:

- For the most routine topics, chat duration falls by about **10%**.
- For moderately rare topics, the drop is closer to **14%**, the largest effect in the distribution.
- For the very rarest topics, the gain is smaller again, around **11%**.
- Holding overall topic frequency fixed, AI helps the most on problems that are rare *for the specific worker*: chat duration falls about **15%** for the least familiar topics compared with about **10%** for the most familiar.
- Figure IX shows clear gains in comprehensibility and native fluency, with especially strong improvements for workers in the Philippines.
- The appendix also shows textual convergence: the similarity between high- and low-skill workers'

(A) Impact of AI on Chat Duration, by Overall Topic Frequency



(B) Impact of AI on Chat Duration, by Agent Topic Frequency

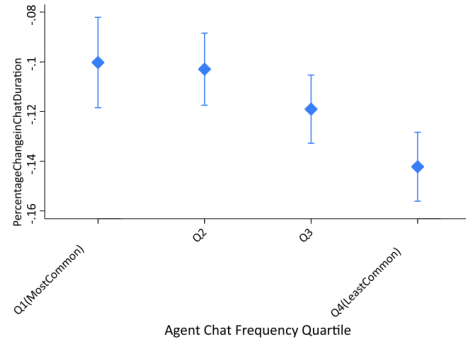


FIGURE VIII

AI Impact by Chat Topic

Panel A shows the impact of AI assistance on chat duration in minutes for each category of conversation-topic frequency relative to the category's pre-AI mean. Data is at the chat level and the regressions control for topic frequency, year-month fixed effects, agent fixed effects, and fixed effects in months of agent tenure. Panel B shows the impact of AI assistance grouped instead by topic frequency as encountered by the individual agent relative to the pre-AI mean. [Online Appendix J.C](#) details the regression specification and topic-category construction and results are in [Online Appendix Table A.IX](#).

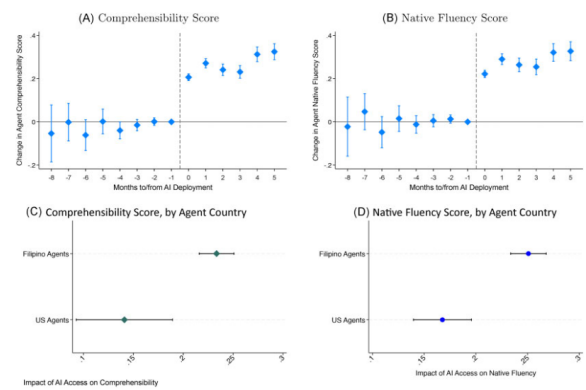


FIGURE IX

The Impact of AI on Language Skills

These figures show the impact of AI access on scores of agent comprehensibility in Panel A and native fluency in Panel B. Observations for this regression are at the agent-chat level, aggregated to the agent-month level. Regressions follow [equation \(1\)](#) and include agent, chat year-month, and months of agent-tenure fixed effects. Robust standard errors are clustered at the agent level in Panels A and B and by agent location in Panels C and D. For more details on the construction of the comprehensibility and native-fluency scores, refer to [Online Appendix J.B](#). Regression results are in [Online Appendix Table A.X](#).

text rises from about **0.55** to **0.61** after AI adoption.

Economic message. AI is most valuable where the machine still knows enough but the human worker does not. That is exactly the pattern one would expect from a knowledge-complementing technology.

5.8 Table IV and Figure X: the experience of work

TABLE IV
EXPERIENCE OF WORK

Variables	Mean customer sentiment (1)	Mean agent sentiment (2)	Share req. manager (3)
Post AI × Ever treated	0.177*** (0.0116)	0.0198*** (0.00599)	−0.00875*** (0.00201)
Observations	21,218	21,218	21,839
R-squared	0.485	0.596	0.482
Year month FE	Yes	Yes	Yes
Agent FE	Yes	Yes	Yes
Agent tenure FE	Yes	Yes	Yes
DV mean	0.141	0.896	0.0377

Notes. This table presents the results of difference-in-difference regressions estimating the effect of AI model deployment on measures of conversation sentiment and requests to speak to a manager (Share req. manager). Our regression specification, equation (1), includes fixed effects for each agent, chat year-month, and months of agent tenure. Observations for these regressions are at the agent-month level and all standard errors are clustered at the agent level. Measures of customer sentiment are created from conversation transcripts using SiBERT and are aggregated to the agent-month level. Online Appendix J.B elaborates on sentiment construction and Section IV.A describes the AI rollout procedure. Robust standard errors are in parentheses. * $p < .10$, ** $p < .05$, *** $p < .01$.

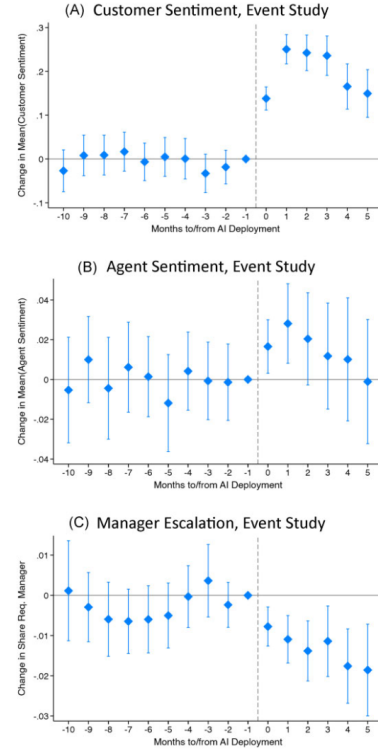


FIGURE X

Experience of Work

Each panel plots the effect of AI model deployment on the experience of work. Panel A plots the effect of AI model deployment on customer sentiment, Panel B plots the corresponding estimate for agent sentiment, and Panel C shows the effects of AI assistance on customer requests for manager assistance. Sentiment is measured using SiBERT, a fine-tuned checkpoint of RoBERTa, an English-language transformer model. Regressions follow equation (1) and include agent, chat year-month, and months of agent-tenure fixed effects. Observations are at the chat level, aggregated to the agent-month, and robust standard errors are clustered at the agent level.

Read:

- AI raises mean customer sentiment by about **0.177**, which the paper describes as about **half a standard deviation**.
- Agent sentiment barely moves, which makes sense because agents already start from a very positive baseline.
- The share of chats in which customers ask to speak to a manager falls by about **0.00875** off a baseline of **0.0377**, roughly a **23%–25%** decline.
- In the appendix, attrition falls most strongly for workers with less than six months of experience:

by around **10 percentage points** relative to a baseline attrition rate of about **25%**.

Economic message. AI changes more than output. It seems to improve customer interactions, reduce escalation pressure, and possibly make the job more sustainable for newer workers.

6 Short summary

- The paper studies the staggered rollout of a GPT-3-based assistant for 5,172 customer-support agents.
- The main design is a worker fixed-effects difference-in-differences model with tenure controls and staggered-adoption event studies.
- AI access raises resolutions per hour by about 15% on average.
- The gain comes mostly from shorter handling times and more chats handled per hour.
- Lower-skill and less-experienced workers benefit much more than top workers.
- AI compresses the experience curve and speeds up worker learning.
- Outage evidence suggests that workers internalize useful practices rather than merely depending on the tool in real time.
- The largest gains occur for moderately rare problems and for topics that are relatively unfamiliar to the specific worker.
- AI improves English fluency, especially for international workers, and makes lower-skill workers communicate more like top performers.
- Customer sentiment improves, manager-escalation requests fall, and novice attrition appears to decline.
- The paper's main message is that generative AI in this setting transfers tacit knowledge from strong workers to weaker workers rather than simply replacing labor outright.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

The paper establishes that a generative-AI assistant can have large and economically meaningful effects in a real production environment. In this customer-support setting, AI substantially raises average productivity, and the gains are highly uneven: they are concentrated among workers who are newer, less experienced, and less skilled at baseline.

7.2 Economic intuition

The most useful way to think about the tool is not as a generic automation device that does part of the job for everyone equally. Instead, it looks like a system that captures successful patterns

from historical conversations—especially the tacit communication and diagnostic habits of top performers—and makes those patterns available to weaker workers in real time. That mechanism naturally implies large gains for workers who have more to learn and small gains for those already close to the frontier.

7.3 Takeaway

The broader takeaway is that generative AI may reshape within-job inequality before it reshapes whole occupations. In the short run, it can flatten productivity differences, reduce training frictions, and improve some dimensions of work quality. But the paper also cautions that these results come from one setting and do not answer longer-run questions about wages, staffing, incentives, equilibrium adjustment, or how AI changes the value of the top workers whose conversations help train the model. Even so, the study is an important benchmark: it shows that generative AI can be a powerful organizational technology for codifying and diffusing know-how.